Ángel Andrés Daruna

Applied ML Research Scientist | Computer Vision | Geospatial AI | Robotics

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Professional Summary

ML researcher with a CVPR-published worldwide visual geolocation system, specializing in ML for computer vision, geospatial intelligence, and robotics. Ph.D.-trained with a track record of delivering AI/ML systems on DARPA- and NGA-funded R&D, including visual geolocation, prospectivity mapping, and uncertainty estimation.

Select Publications *(full list available upon request)*

1. GeoSURGE: Geo-localization using Semantic Fusion with Hierarchy of Geographic Embeddings." **CVPR 2026.** https://arxiv.org/abs/2510.01448
2. "Ground-to-Satellite Localization in Unconstrained Image Collections for 3D Scene Reconstruction." **IGARSS 2026**.
3. "Enabling Scalable Mineral Exploration: Self-Supervision and Explainability." **IEEE Big Data 2024.** https://ieeexplore.ieee.org/abstract/document/10825956/
4. "GFM4MPM: Towards Geospatial Foundation Models for Mineral Prospectivity Mapping." **ACM SIGSPATIAL 2024.** https://dl.acm.org/doi/abs/10.1145/3678717.3691268
5. "Explainable Knowledge Graph Embedding: Inference Reconciliation for Knowledge Inferences Supporting Robot Actions." **IEEE IROS 2022.** https://ieeexplore.ieee.org/abstract/document/9982104

Skills

Software: Python, PyTorch, PyTorch Lightning, CUDA, Docker, AWS, ROS

Machine Learning: self-supervised learning, contrastive learning, foundation models, explainable AI, RAG

Computer Vision: multimodal fusion, visual localization, cross-view matching, 3D vision

Education

Georgia Institute of Technology Atlanta, GA

Ph.D. in Robotics, 3.76 GPA, 2022; M.S. in CS, 3.76 GPA, 2021; B.S. in ComEng 3.91 GPA, 2015

Recent Experience

Advanced Computer Scientist @ SRI International

2021 – Present | Remote, FL

- Led development of a large-scale visual geolocation system combining semantic segmentation with vision transformers using cross-attention, trained on 15M+ images across 8 A6000 Nvidia GPUs; achieved state-of-the-art in 22 of 25 metrics measured across 5 benchmark datasets. Published at CVPR 2026.
- Designed mineral exploration system enabling geologists to use diverse geospatial data (remote sensing, geophysics, geology, geochemistry) to train vision transformers using self-supervised learning; consistently improving over prior methods for a suite of metrics (e.g., F1-score +5%), being robust to highly imbalanced, scarce, or missing training data; deployed to USGS infrastructure via Docker and AWS S3/EC2.
- Secured extended funding across 7 government contracts (DARPA, NGA, USGS, IARPA) by consistently delivering AI/ML systems that outperformed during down selection, contributing to 2-3 concurrent projects; co-authored 4 conference articles, 3 patent applications, and 2 proposals funded totaling over $2M.

Graduate Researcher @ Georgia Tech

2016 – 2022 | Atlanta, GA

Advisor: Sonia Chernova

- Led integration of point-to-point navigation and collision mapping; developed a ROS package for accurate object pose estimation (pen-and-hole at 3 mm clearance) on a robot architecture that won the IEEE ICRA 2019 FetchIt! Mobile Manipulation Challenge operating autonomously for 45 minutes (kit assembly).
- Explored knowledge graph embedding for robotics, enabling robots to infer missing commonsense knowledge. Extended this foundation to support continual learning and natural language explanations of inferred knowledge. Published 12 articles over a diverse, collaborative Ph.D research program.